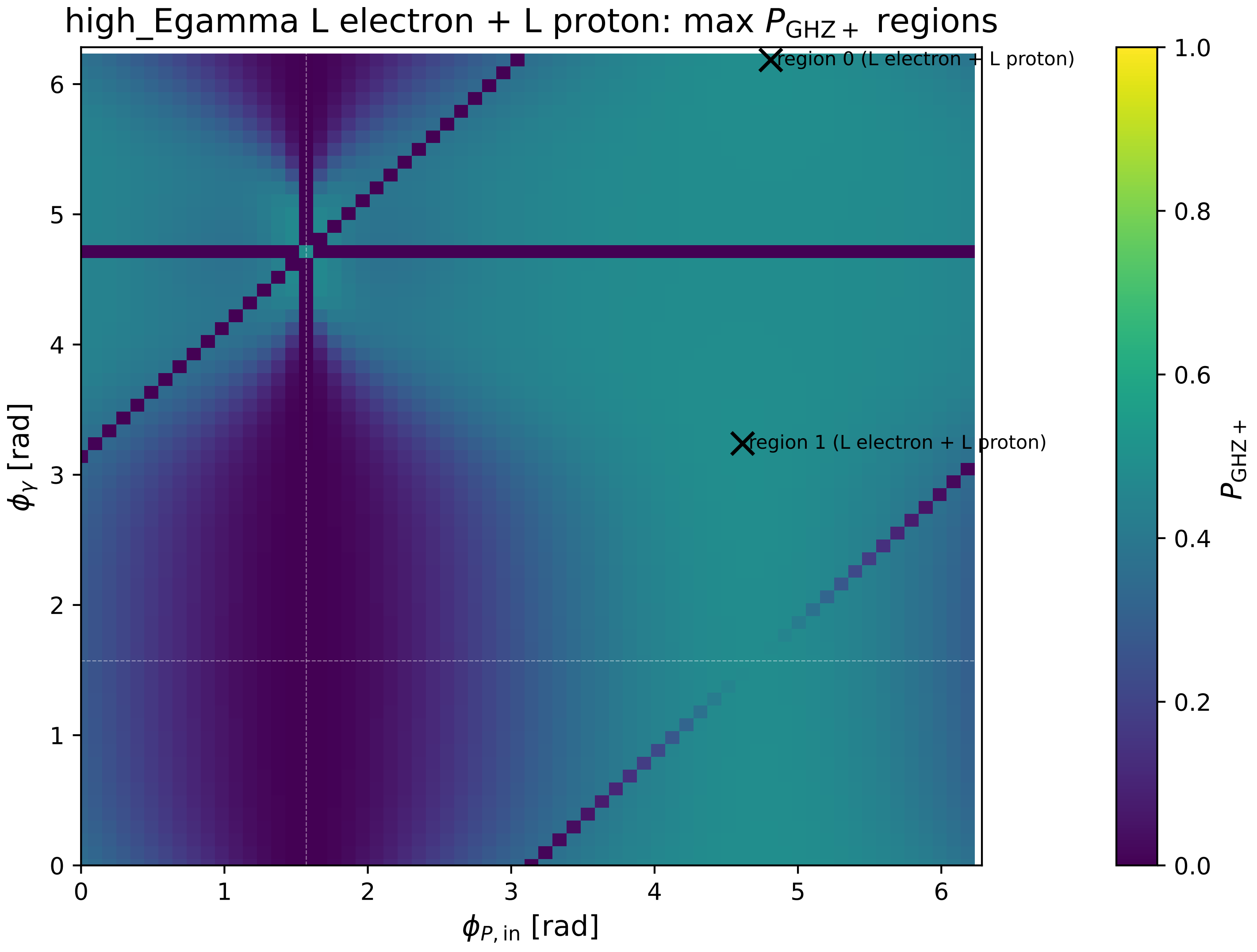
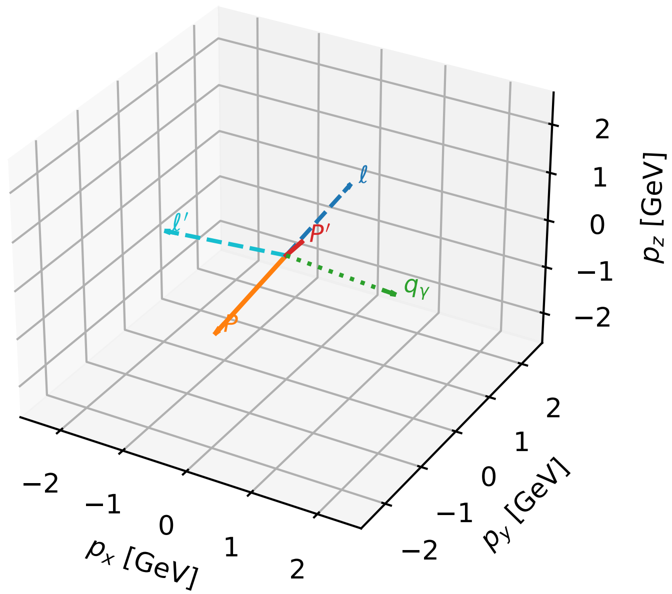

high_Egamma L electron + L proton: max $P_{\mathrm{GHZ}+}$ regions
region 0 (L electron + L proton)
region 1 (L electron + L proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$P_{\mathrm{GHZ}+}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

L electron + L proton characteristic max $P_{\rm GHZ+}$ configuration

3D momenta

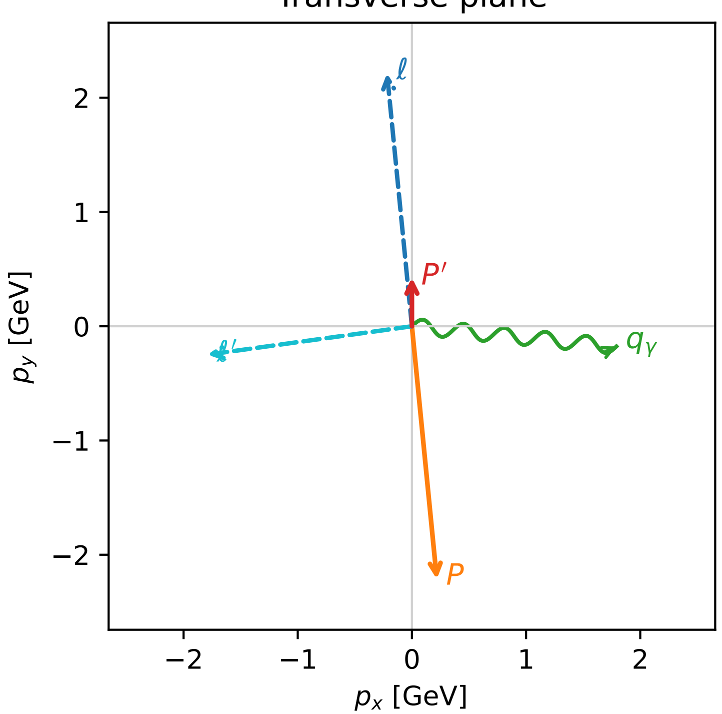



```
ghz_purity_high_egamma_ll_region_0  P_{\rm GHZ+}=0.489608
region: high_Egamma_LL_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
```
incoming state: $|L_e\rangle \otimes |L_p\rangle$

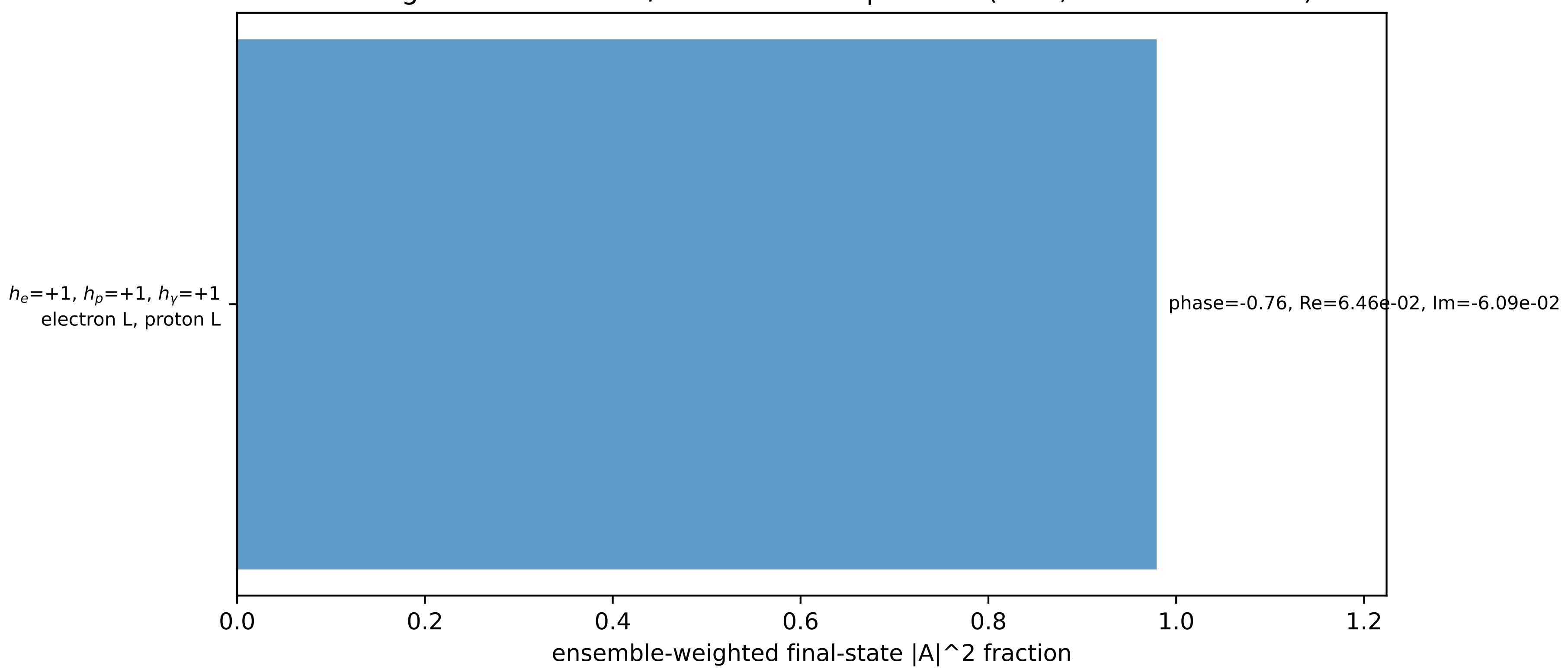
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.425418
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.66897, ϕ_P=4.81056
E_γ=1.8, ϕ_γ=6.18501
Q^2=8.3672, x_B=0.684423, t=-5.09668, W^2=4.73783, y=0.59242
$\theta(\ell', \gamma)$=166.462 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -0.2180 p_y= 2.2137 p_z= -0.0000
P E= 2.4141 p_x= 0.2180 p_y= -2.2137 p_z= 0.0000
ℓ' E= 1.8086 p_x= -1.7913 p_y= -0.2490 p_z= 0.0000
P' E= 1.0300 p_x= 0.0000 p_y= 0.4254 p_z= 0.0000
q_γ E= 1.8000 p_x= 1.7913 p_y= -0.1764 p_z= 0.0000

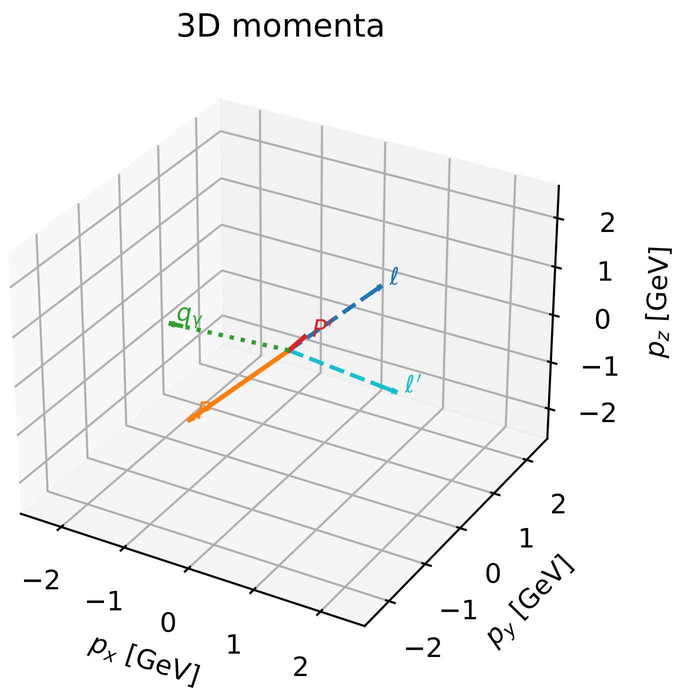
Transverse plane



Leading initial-ensemble/final-state components (N=1, retained=97.9%)



L electron + L proton characteristic max $P_{\rm GHZ+}$ configuration



ghz_purity_high_egamma_ll_region_1 P_{\rm GHZ+}=0.489608
region: high_Egamma_LL_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_e\rangle \otimes |L_p\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.425418
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.47262, ϕ_P=4.61421
E_γ=1.8, ϕ_γ=3.23977
Q^2=8.3672, x_B=0.684423, t=-5.09668, W^2=4.73783, y=0.59242
$\theta(\ell',\gamma)$=166.462 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.2180 p_y= 2.2137 p_z= -0.0000
P E= 2.4141 p_x= -0.2180 p_y= -2.2137 p_z= 0.0000
ℓ' E= 1.8086 p_x= 1.7913 p_y= -0.2490 p_z= 0.0000
P' E= 1.0300 p_x= 0.0000 p_y= 0.4254 p_z= 0.0000
q_γ E= 1.8000 p_x= -1.7913 p_y= -0.1764 p_z= 0.0000